Scientific Analysis Group (SAG), DRDO

Internship Progress Report

Flow-Pattern Anomaly Detection at the IoT Gateway

A Pre-trained Mamba Detector for Known and Zero-Day Attacks

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`flowmamba` prototype · IoT-23 real malware traffic · CPU (Apple M3 Pro)

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Abstract

IoT gateways see every device flow but cannot host endpoint security agents, so attack detection must happen *on the network*, cheaply, and on encrypted traffic. This report documents `flowmamba`, a lightweight detector that keeps the *packet order* of a connection — the first K packets as an ordered sequence of payload-free features — and feeds it to a shared Mamba state-space encoder with two heads: a supervised classifier for known attacks and a one-class deep-SVDD scorer for zero-days. We trace the empirical journey from a single-capture prototype (macro-F1 0.79, anomaly ROC-AUC 0.994) to a multi-source model trained on seven IoT-23 captures. The central practical finding is that, with payload-free features on *short* flows, the model separates traffic by **flow structure** — benign vs. short scan/botnet attacks vs. long brute-force — but *not* by fine attack family (DDoS, Recon and Mirai-C&C are inseparable in the first few packets, proven three independent ways). Aligning the label taxonomy with what the features can actually resolve yields a clean three-class detector at **98.2 % accuracy**, macro-F1 **0.984**, anomaly ROC-AUC **0.935**, and ∼5 ms/flow latency. We close with the open question a CNN sub-classifier is meant to test.

Headline. Final three-class model — **98.2 % accuracy**, macro-F1 **0.984**, anomaly ROC-AUC **0.935**, 5.0 ms/flow — trained on seven real IoT-23 malware captures. The accuracy ceiling is set by genuine feature overlap among short-flow attack families, not by model capacity.

1 Motivation

The target environment is the modern smart home, escalating to a defence context: *"field sensors, surveillance equipment and base infrastructure are high-value, low-trust devices that cannot be hardened one by one."* IoT devices ship with default credentials, run rarely-updated firmware, and lack the compute to host any endpoint agent — the Mirai botnet, built from compromised IP cameras and routers, is the canonical proof this surface is exploitable at scale. IoT anomaly detection at large is surveyed in [19, 20].

Since software cannot sit *on* the devices, the detector watches them *on the network*. The gateway is the single point through which every device flow passes, and the only place modern compute can sit without instrumenting the fleet. That vantage comes with one hard constraint that shapes every design choice: gateway hardware is Raspberry-Pi-class, so the detector must be small and fast — a target of **<10 ms per flow**.

A useful gateway detector must do four things at once:

1. **Catch known attacks** with high precision (supervised classifier).
2. **Catch unknown, zero-day attacks** — learn "normal" and flag whatever does not fit, *without ever training on the attack* (one-class anomaly head).
3. **Work on encrypted traffic** — read flow metadata only, never payload.
4. **Run at the gateway** on small hardware, under 10 ms per flow.

2 Related Work

IoT anomaly detection is a crowded field; recent surveys map the landscape [19, 20]. Four families are relevant here, and each meets only part of the gateway brief (Table 1).

Signature / firewall rules catch known attacks precisely but miss zero-days and rely on deep packet inspection, which encryption defeats. **Flow-statistic ML** (XGBoost, random forests on ≈80 aggregate features) is cheap and edge-friendly but *order-agnostic* — it discards the packet-sequence structure that attack signatures live in. **Pre-trained transformers** (ET-BERT [5]) reach state-of-the-art accuracy on encrypted traffic but at ∼110M parameters do not fit gateway hardware; even purpose-built transformer studies for constrained IoT report this tension [21].

One-class / autoencoder detectors catch novelty but cannot name known-attack categories.

Our design follows NetMamba [1]: a selective state-space encoder that keeps transformer-class accuracy at edge cost, extended with a *shared encoder + dual head* so one representation serves both known-attack classification and zero-day anomaly scoring. Complementary directions — graph IDS (E-GraphSAGE [18]), federated training across sites [24, 25], and explainable zero-day alerts [22] — are noted as future extensions.

Table 1: Where existing approaches meet the four gateway requirements.

Approach	Known	Zero-day	Encrypted	Edge
Signature / firewall	✓	—	—	✓
Flow-statistic ML (XGBoost)	✓	∼	✓	✓
Transformer (ET-BERT)	✓	∼	✓	—
One-class / autoencoder	—	✓	✓	∼
This work (Mamba dual-head)	✓	✓	✓	✓

3 Approach and Architecture

3.1 The representational bet: keep the order

Conventional IDS collapses a flow into ∼80 summary statistics — convenient but *permutation-invariant*: shuffle the packets and the statistics do not change, destroying the temporal structure attack signatures live in. The core bet of this work is the opposite: **keep the order**. A flow is the first $K \approx 32$ packets of a connection, kept as an ordered sequence $x = (p_1, \dots, p_K)$, each packet described by a 13-dimensional payload-free vector (Table 2). Because the features are header/metadata only, the representation works on encrypted traffic.

Table 2: The 13 payload-free per-packet features (`flowmamba.data.features`).

Group	Features
Size / timing	`size` (bytes), `iat` (inter-arrival time), `header_len`
Direction	`direction` (+1 up / −1 down)
TCP flags	`syn`, `ack`, `fin`, `rst`, `psh`, `urg`
Header fields	`ttl`, `win` (TCP window), `is_tcp`

3.2 Why Mamba, and the shared-encoder design

The reference pre-trained traffic detector, ET-BERT [5], is a ∼110M-parameter transformer that does not fit on a Raspberry Pi [21]. NetMamba [1] showed a Mamba [2] state-space model reaches comparable accuracy at far lower parameter and latency cost — Mamba is $O(K)$ in sequence length with $O(1)$ inference memory, versus attention's $O(K^2)$; the selective state-space lineage runs from S4D [4] through Mamba to Mamba-2 [3]. A *single* encoder is shared by both heads (Fig. 1): pre-training is the expensive part, so sharing pays it once, and it hands the anomaly head a representation of "normal" learned from a large benign corpus — exactly what a one-class scorer needs. The guiding principle: *"what's normal" lives in the expensive encoder; "what's malicious" lives in the cheap heads.*

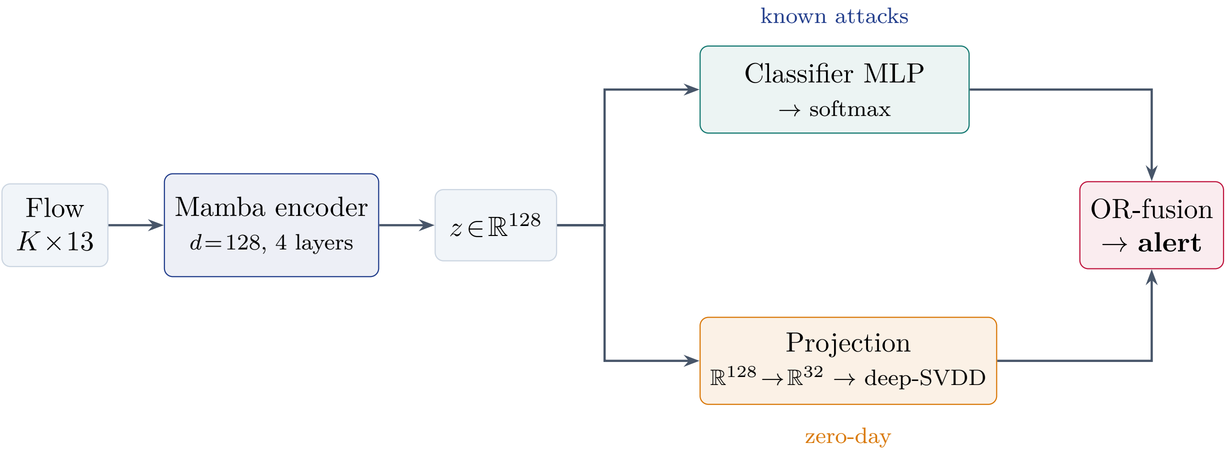


Figure 1: Shared Mamba encoder with dual heads. The classifier names known attack categories; the benign-trained deep-SVDD head flags anything far from "normal". Either head crossing its threshold raises an alert.

3.3 Three-stage training

Training decouples the expensive representation from the cheap decision heads (Fig. 2):

- **Stage A — SSL masked pre-training** (benign only, no labels). Mask ∼15% of packet positions; train the encoder to in-fill them from context (MSE on masked positions); a contrastive objective over augmented flow views (ET-SSL [6]) is an ablatable alternative. Output: encoder θ_0 that has learned normal traffic.
- **Stage B — supervised fine-tune.** Attach a fresh classifier on θ_0; jointly fine-tune with focal loss [8] ($\gamma = 2$) on labelled attacks. Output: $\theta_\star$ + classifier.
- **Stage C — deep-SVDD [7] anomaly head.** Freeze $\theta_\star$; push a held-out benign set through it; train the projection MLP under the deep-SVDD objective; calibrate the radius at the **99th benign percentile (1% FPR target)**.

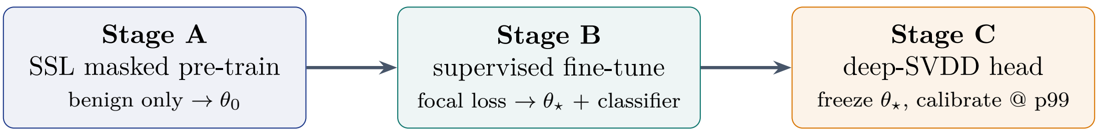


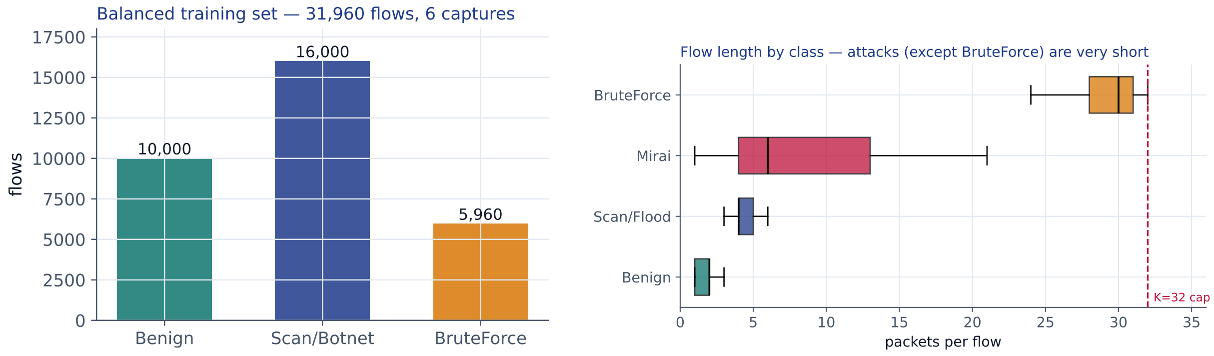
Figure 2: Three-stage pipeline. Self-supervision learns "normal" once; the two heads are trained cheaply on top of the frozen/fine-tuned encoder.

Two operating modes share these same artifacts: a low-power *default* mode (smaller K, anomaly head only, target <2 ms) and a full-pipeline *strong* mode ($K = 32$, both heads, per-alert SHAP [11] attributions, target <10 ms).

4 Data and Feature Representation

Flows are extracted from IoT-23 [9] (Stratosphere Labs) packet captures with `dpkt`; labels are joined from each Zeek `conn.log.labeled` by 5-tuple. Seven captures were used — two from the initial prototype (`34-1` Mirai, `8-1`) and five added for this report (`1-1`, `3-1`, `20-1`, `21-1`, `42-1`) — chosen for attack-type diversity and manageable size. IoT-23 detailed labels map onto the model's categories as: `PartOfAHorizontalPortScan`→Recon, `DDoS`→DDoS, `Attack`→BruteForce, and all C&C/Okiru/Torii→Mirai. Per-class flows are *water-filled* across source captures (so each class draws from several scenarios) and balanced into a 31,960-flow training set (Fig. 3, left).

The single most consequential property of this data is **flow length** (Fig. 3, right): benign connections and short scan/flood attacks are tiny (∼2–7 packets, median ≤6), while brute-force flows are long (∼30 packets). Over 99% of the short-flow classes never reach the $K = 32$ cap — a fact that decides what any sequence model can and cannot separate.



(a) Balanced training set (3 classes). **(b)** Flow length by class — the decisive property.

Figure 3: Left: the class-balanced multi-capture dataset. Right: attacks (except brute-force) are short connections; there is little sequence for a model to work with.

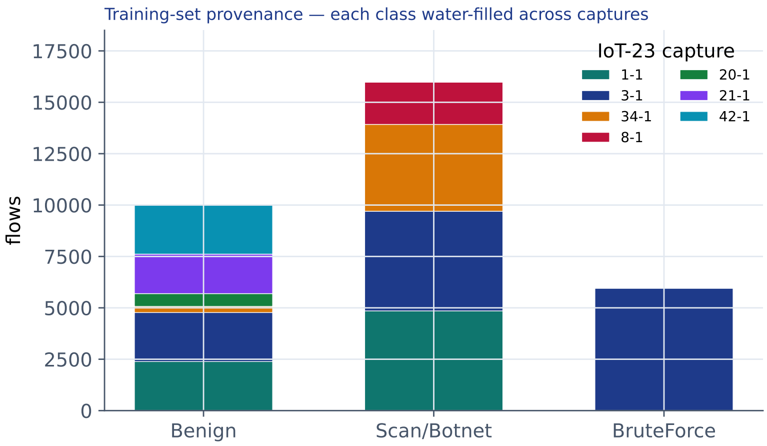


Figure 4: Each class is *water-filled* across the seven captures so no single scenario dominates — Benign is drawn roughly evenly from six captures, which broadens the notion of "normal" the anomaly head learns.

5 Feature Significance and Preprocessing

5.1 Taming heavy-tailed features

Raw traffic features — packet sizes, byte counts, inter-arrival times — are heavy-tailed, spanning orders of magnitude; a single bulk transfer would dominate any distance-based score. We compress the tail with a $\log(1+x)$ / Yeo–Johnson [14] power transform, then standardise (Fig. 5). The direction matters: log compacts the right tail, exp would stretch it the wrong way.

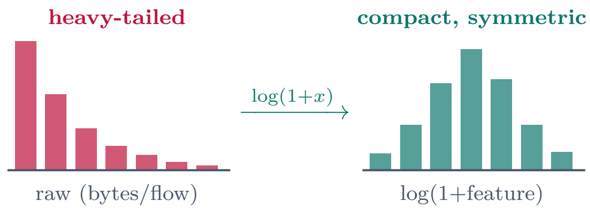


Figure 5: The log/Yeo–Johnson transform pulls the long right tail (rose) into a compact, near-symmetric distribution (amber) so distances reflect the bulk of the data, not a few outliers.

5.2 Ranking, then confirming

Feature *significance* for the classical baselines is decided by mutual information and SHAP [11]; t-SNE [12] / UMAP [13] only *confirm* separability — they are visualisation tools, not rankers (their axes are nonlinear blends of all inputs). For the Mamba arm there is no hand-picked vector: the encoder learns its own embedding z, and the right check is a t-SNE of z (Fig. 6).

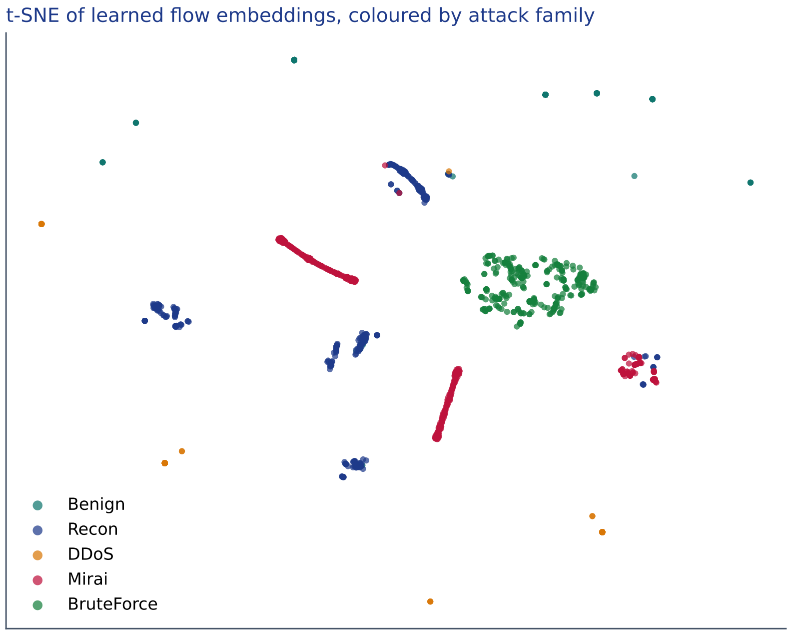


Figure 6: t-SNE of the learned flow embeddings (1,411 flows). **BruteForce** (long, ∼30-packet) and **Mirai** form distinct clusters; **Recon** fragments into several small clusters; **DDoS** (only 211 flows) is too sparse to place. A caution: t-SNE's 2-D proximity is *not* classifier separability — a capture-controlled pairwise test (Table 6) shows Mirai is *fully* separable from the scan classes, while Recon↔DDoS is the genuinely fused pair.

5.3 The anomaly sphere

The zero-day head is a deep-SVDD [7] one-class scorer: a learned projection $\mathbb{R}^{128} \to \mathbb{R}^{32}$ packs benign embeddings into a minimum-volume ball around a centre c; a flow beyond radius R (99th benign percentile, 1% FPR) is flagged (Fig. 7). An L2-normalised angular-cap variant is a clean A/B alternative. In practice the benign scores pack tightly near the centre while attacks spread out (Fig. 7).

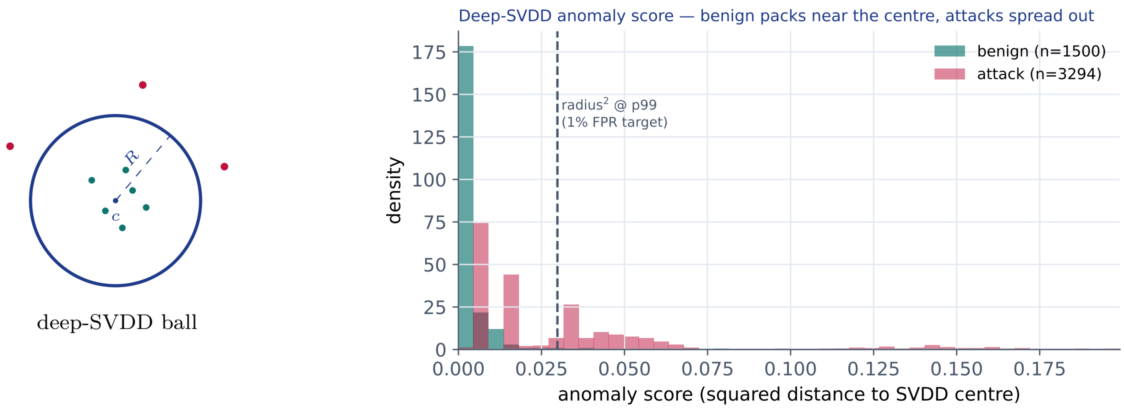


Figure 7: Left: deep-SVDD packs benign embeddings (amber) into a ball of radius R around c; anomalies (rose) fall outside. Right: the measured score distribution — benign concentrates near zero, attacks spread past the calibrated radius.

6 Experimental Journey

The project evolved through a sequence of experiments, each of which produced a concrete finding rather than just a number (Table 3, Fig. 8).

Table 3: The experimental journey. Macro-F1 is over the classes present in each setup.

Experiment	Acc	Macro-F1	Anom.	Finding
34-1, 4-class (random split)	0.963	0.793	0.994	Anomaly head excellent; DDoS↔Recon confused
34-1 + stratify + focal	–	0.77 / 0.67	0.99	Reweighting *see-saws* ⇒ genuine overlap
34-1 merged, 3-class	0.991	**0.957**	0.994	Merging DDoS+Recon → Scan/Flood fixes it
34-1+8-1 combined, 4-class	–	0.686	0.991	Pure-Mirai capture *regressed* the classifier
Multi-source, 4-class	0.906	0.895	0.913	New BruteForce class (F1 0.996); Mirai recall 0.60
Multi-source, Mirai↑2.5×	0.830	0.832	–	See-saw again ⇒ Mirai/Scan overlap
Multi-source, 3-class (final)	**0.982**	**0.984**	**0.935**	Align taxonomy with separability

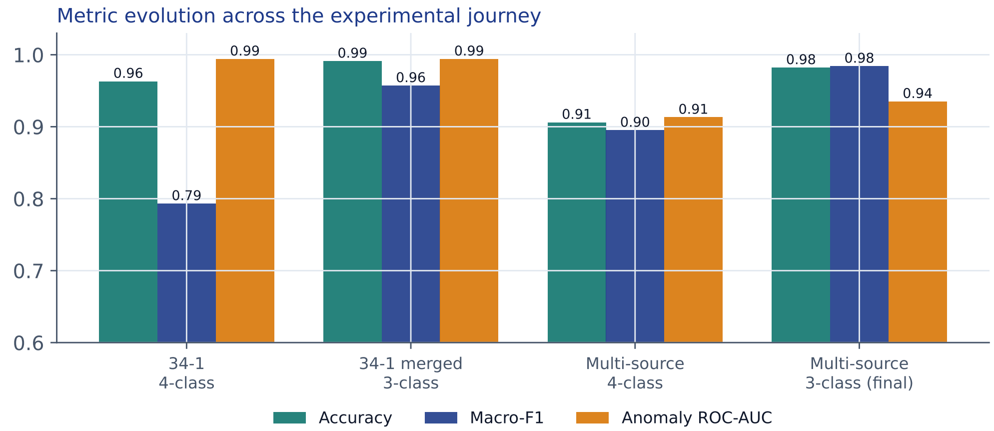


Figure 8: Metric evolution across the four milestone models. Numbers span evolving class sets and test splits, so they read as a trajectory rather than a strict apples-to-apples comparison.

6.1 From one capture to many

The prototype on a single Mirai capture (`34-1`) already validated the central bet — the benign-trained anomaly head hit **ROC-AUC 0.994** on real malware, at sub-6 ms latency. But the supervised classifier persistently confused DDoS with Recon. Two attempted fixes (stratified splits, per-class focal weights) only made the errors *swap direction*: upweighting one class sacrificed the other. This *see-saw* is the signature of genuine feature overlap, not class imbalance. Merging the two indistinguishable classes into a single **Scan/Flood** class resolved it cleanly (macro-F1 0.957).

Scaling to more captures added a genuinely new, well-separated class — **BruteForce** (F1 0.996) — and a much more diverse benign set, at the cost of a lower but *more honest* anomaly ROC-AUC (0.913 vs. the inflated single-capture 0.994). It also surfaced the same pathology one level up: the now-separate **Mirai** class was confused with Scan/Flood (recall 0.60).

6.2 The see-saw, and why longer sequences cannot help

Upweighting Mirai 2.5× drove its recall to 0.999 — but collapsed Scan/Flood recall to 0.503 and dropped overall accuracy to 0.830 (Fig. 9). The confusion did not vanish; it flipped and grew. The obvious next idea — longer sequences ($K = 64$) to let persistent C&C diverge from short scans — was ruled out for *free* by the length analysis: over 99% of these flows never reach even 32 packets, so there are no extra packets to use. Mirai botnet traffic *is* partly horizontal port-scanning; over the first ~6 packets it is genuinely indistinguishable from the scan class.

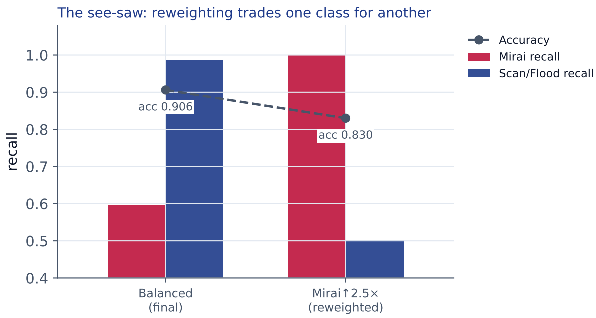


Figure 9: Reweighting is a zero-sum trade between two overlapping classes: Mirai recall rises only as Scan/Flood recall falls, and accuracy drops.

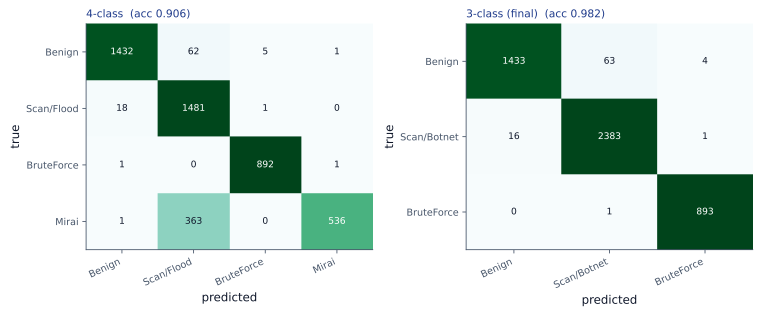
7 Final Model and Results

Aligning the taxonomy with what the features can resolve — Benign, a combined **Scan/Botnet** class (DDoS, Recon and Mirai, all short-flow attacks), and **BruteForce** — gives a clean, high-accuracy detector. Trained for 3 stages × 6 epochs on 31,960 flows with a stratified split and mild $\sqrt{}$-frequency focal weights, it reaches **98.2 % accuracy** (Table 4, Fig. 10).

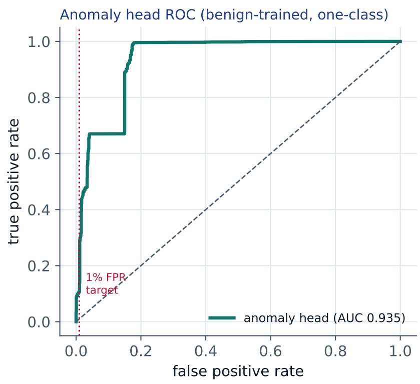
Table 4: Final three-class model on the held-out test set (4,794 flows).

Class	Precision	Recall	F1	N
Benign	0.989	0.955	0.972	1500
Scan/Botnet	0.974	0.993	0.983	2400
BruteForce	0.994	0.999	**0.997**	894
Macro avg	0.986	0.982	**0.984**	4794
Weighted avg	0.982	0.982	0.982	4794

Metric	Value
Accuracy	**98.2%**
Anomaly ROC-AUC	**0.935**
Anomaly PR-AUC	0.959
Latency p50/p95	5.0 / 6.5 ms
Model size	2.0 MB



(a) Row-normalised confusion: 4-class (left) vs. final 3-class (right).



(b) Anomaly-head ROC.

Figure 10: Left: merging the inseparable short-flow attacks turns the 4-class Mirai→Scan smear into a near-diagonal 3-class matrix. Right: the one-class anomaly head at ROC-AUC 0.935 on hard, multi-source data.

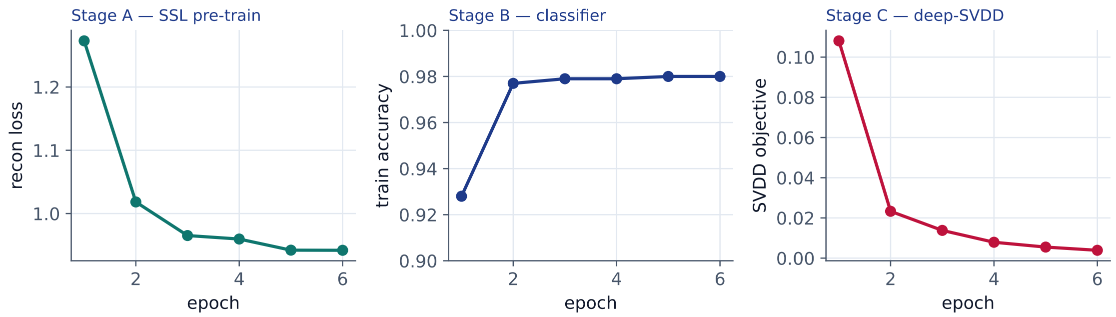


Figure 11: Training convergence (final 3-class run). All three stages plateau well inside 6 epochs; the accuracy ceiling is data/feature-bound, not under-training.

8 Deployment and Quantization

The detector must fit on Raspberry-Pi-class hardware, so the deployed model is *quantised* — weights and activations stored in fewer bits than fp32. The wrinkle is that most quantisation work targets transformers, whereas Mamba's *selective state-space core* carries activation outliers that compound through its recurrence and resist naive int8.

Strategy. *Mixed precision* — int8 for the linear projections (which dominate the parameter count and quantise cleanly), fp16 for the outlier-sensitive selective-scan core (Fig. 12). Most of the size saving comes from the projections anyway; quantisation-aware training is the fallback if fp16 still costs accuracy on the SSM core [15, 16, 17].

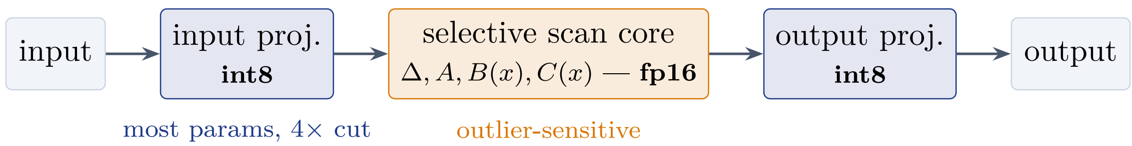


Figure 12: Mixed-precision split inside a Mamba block: int8 projections (which hold most weights), fp16 selective-scan core — the compromise across the 2024–25 Mamba quantisation literature (Q-Mamba, MambaQuant, PTQ4SSM).

The plan reports three precisions so the operator can trade size against accuracy (Table 5). **These are design projections, not measured** — the shipped `detector.pt` is the 2 MB fp32 model, and the int8 conversion plus the on-device (Raspberry-Pi) benchmark remain outstanding (§9).

Table 5: Planned quantisation configurations (projected, per the deployment design).

Configuration	Size	p95 latency	Accuracy hit
fp32 (baseline)	1.0×	1.0×	—
int8 PTQ everywhere	0.25×	∼0.5×	risk on SSM core
Mixed precision (recommended)	≈ 0.30×	∼0.55×	smallest drop

9 Continual and Progressive Learning

Attack landscapes drift, so a deployed detector must absorb new attack variants without forgetting old ones. The hard case is *correlation*: a new DDoS variant that resembles standard DDoS. Three failure modes lurk — catastrophic forgetting, probability-mass theft (a new class stealing confidence from old ones), and spurious hard labels on ambiguous flows. The update protocol (Fig. 13) keeps the expensive encoder *frozen* and touches only the cheap heads, with four defences in order: (1) a replay buffer of ∼100–500 flows per existing class; (2) knowledge distillation (learning-without-forgetting) against the previous head; (3) an embedding-space check that decides *add* (tight and far → new class), *merge* (overlapping → sub-class), or *separate* (scattered → margin fine-tune); and (4) an automatic rollback if any existing class's F1 regresses beyond an agreed threshold.

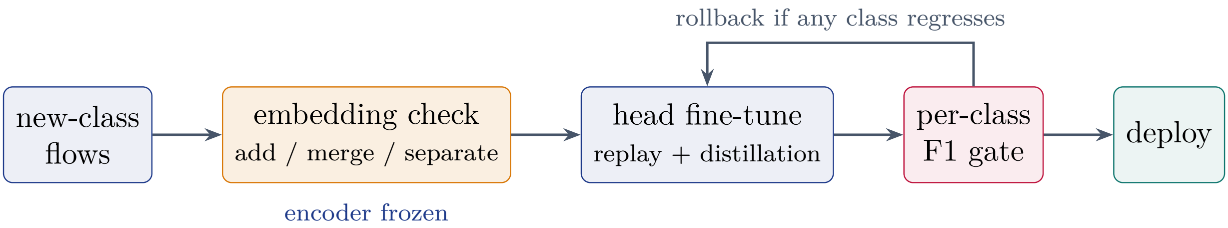


Figure 13: The progressive-learning update: only the cheap heads change; a replay + distillation fine-tune is gated by a per-class F1 check, with automatic rollback.

10 Analysis: what the features can and cannot resolve

The governing result of the whole journey is a single structural fact:

Design finding. Payload-free features on short flows separate traffic by *flow structure* (benign ∼3 pkts, short scan/botnet ∼5 pkts, long brute-force ∼30 pkts) more than by fine attack family. A capture-controlled pairwise test is precise (Table 6): **Mirai is fully separable** from the scan classes, but **Recon↔DDoS is near-chance** — both are short horizontal-scan/flood patterns overlapping in the first ∼6 packets. So the genuinely inseparable pair is Recon/DDoS; Mirai is merged in the *flat* 3-class model only for robustness against cross-capture variety, and the hierarchical CNN layer (§11) recovers it.

Table 6: Pairwise separability within a single capture (34-1), so the score reflects attack behaviour, not capture identity. Random-forest 5-fold accuracy.

Pair	5-fold accuracy	Verdict
Recon vs DDoS	0.736	near-chance — *fused*
Recon vs Mirai	**1.000**	fully separable
DDoS vs Mirai	**1.000**	fully separable

This is not a failure of the model but an honest measurement of the information content of the representation. It has two direct consequences: (i) the **three-class taxonomy** is the right granularity for these features, and (ii) recovering the finer attack families would require information the current input lacks — *flow-level metadata* (duration, byte counts, destination-port diversity, connection state), not more packets. The anomaly head — the project's zero-day centrepiece — is unaffected by this classifier question and remains strong (ROC-AUC 0.935) on diverse, realistic traffic; its earlier 0.994 was inflated by a single Mirai-dominated capture. *Tooling:* heavy-tailed features are Yeo–Johnson [14] transformed before standardisation; embedding separability is verified with t-SNE [12] / UMAP [13]; and a per-window device-communication graph branch (E-GraphSAGE [18]) is a planned extension.

11 A CNN Fine-Layer: Sub-Classifying Scan/Botnet

The Scan/Botnet class deliberately fuses three attack families (DDoS, Recon, Mirai). A natural — and suggested — extension is to leave the detector intact and add a *CNN as a dedicated second-stage layer* whose only job is to split that class back into its families (Fig. 14). We evaluate this two ways.

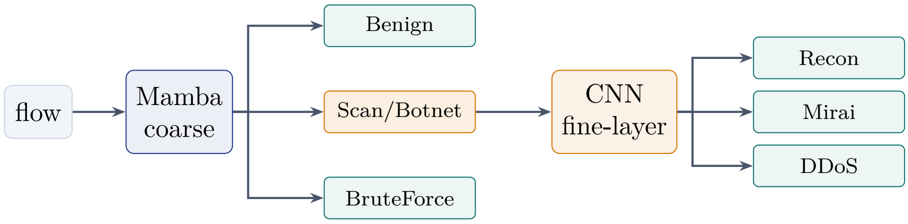


Figure 14: Hierarchical design. The robust coarse detector routes each flow; a CNN fine-layer, trained only on Scan/Botnet flows, names the attack family.

11.1 Does the architecture matter? CNN vs. Mamba vs. XGBoost

Training a CNN, our Mamba encoder, and an order-agnostic XGBoost on the identical three-way task (Recon/DDoS/Mirai) gives a clear verdict (Fig. 15):

- The **CNN matches but does not beat the Mamba** — on both settings they converge to identical predictions (differing only in confidence). The useful idea is the *hierarchy*, not the architecture.
- Both sequence models **beat order-agnostic XGBoost** (0.75 vs. 0.60, single-capture), so keeping packet order helps the fine split too.
- XGBoost's *perfect* 1.00 on multi-capture data was the tell for **capture leakage**: because each family is sourced from different captures, a model can fingerprint the *capture* (via TTL, window size) instead of the attack. The capture-controlled numbers (∼0.75) are the honest ones.

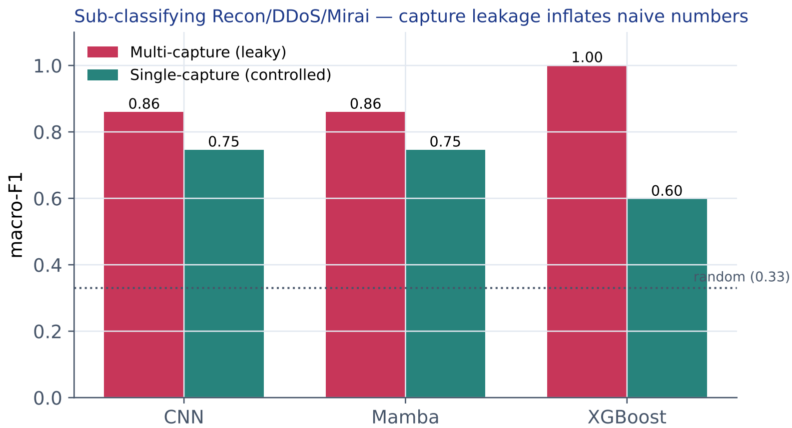


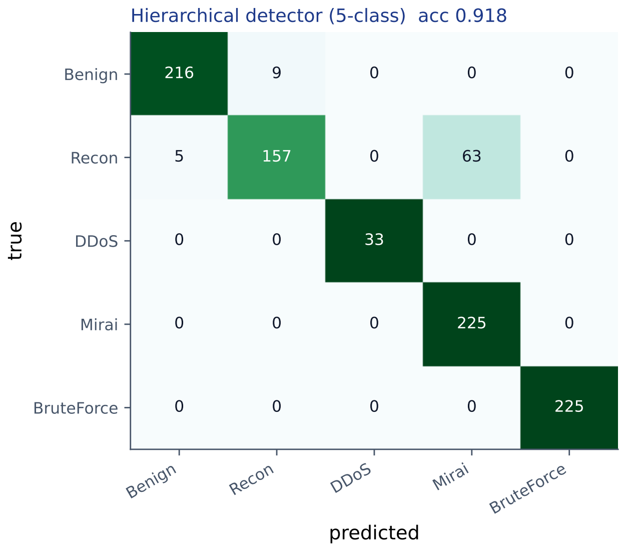
Figure 15: Fine-split macro-F1. All three models drop sharply from multi-capture to single-capture — especially XGBoost (1.00→0.60) — exposing how much of the naive score was capture identity, not attack behaviour.

11.2 The end-to-end hierarchical detector

Wiring the CNN fine-layer beneath the coarse Mamba upgrades the three-class detector into a five-class one that *names* the attack family (Table 16a, Fig. 16). It reaches **91.8 % accuracy** and cleanly recovers DDoS, Mirai and BruteForce; Recon is the weak point (some flows read as Mirai). The flat coarse model alone scores 0.985 on the 3-way but cannot name families — the fine-layer is what buys that capability.

Class	P	R	F1
Benign	0.977	0.960	0.969
Recon	0.946	0.698	0.803
DDoS	1.000	1.000	1.000
Mirai	0.781	1.000	0.877
BruteForce	1.000	1.000	1.000
Accuracy **0.918**	Macro-F1 **0.930**		

(a) Per-class (5-way).



(b) End-to-end confusion.

Figure 16: The hierarchical detector names four attack states plus benign at 91.8 % accuracy; the residual is Recon read as Mirai.

Honesty caveat. These multi-capture fine numbers are inflated by capture leakage: DDoS's perfect score reflects its unique source capture, and the confusion pattern *shifts* with capture composition (single-capture DDoS→Recon; multi-capture Recon→Mirai) — the fingerprint of confounding. The trustworthy fine-split ceiling is ∼0.75 macro-F1, with DDoS↔Recon the genuine limit. A leave-one-capture-out evaluation is the deployment-honest next step.

12 Evaluation Design

Beyond the results reported above, the full evaluation plan pits the detector against four baselines, each isolating a different question (Table 7), plus two stress tests: **FGSM** adversarial perturbations clipped to valid packet ranges [23], and a **leave-one-class-out** zero-day proxy — hide an attack class during supervised training, then measure how well the benign-trained anomaly head still flags it. Headline metrics are per-class P/R/F1, PR-AUC (the honest metric under imbalance), latency p50/p95/p99, and int8 model size.

Table 7: The four baselines, and the question each isolates.

Baseline	Role
ET-BERT [5]	published-SOTA *upper bound* (accuracy ceiling)
CNN-BiLSTM on aggregates	*representation control* — does keeping packet order help?
XGBoost on aggregates	classical *accuracy floor*
Isolation Forest	unsupervised *anomaly floor* for the zero-day head

The deep detector earns its complexity only if it beats the classical floor on zero-day detection, cross-dataset transfer, and adversarial robustness; where XGBoost matches in-distribution, that is reported as an honest finding.

13 Project Timeline

The eight-week internship (DOJ 11$^{\text{th}}$ May 2026) progressed from data plumbing to a validated multi-source detector (Fig. 17).

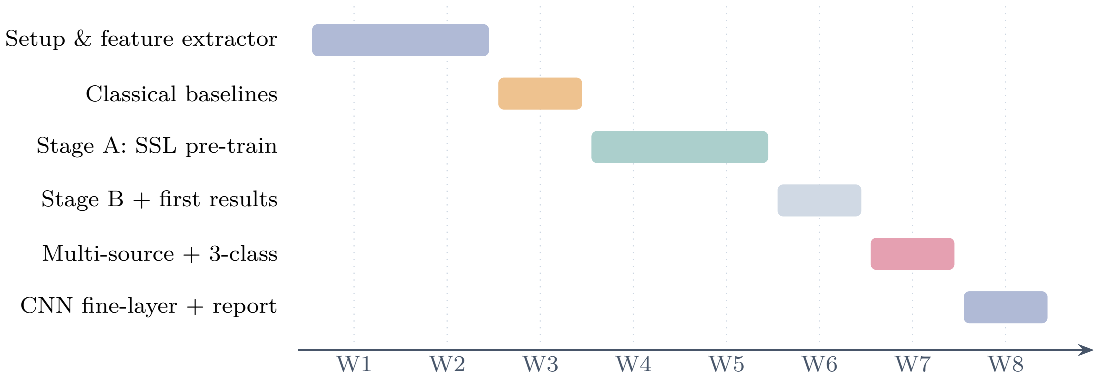


Figure 17: Eight-week timeline: setup and extraction (W1–2), classical baselines (W3), self-supervised pre-training (W4–5), supervised fine-tune and first single-capture results (W6), multi-source data with the 3-class taxonomy (W7), and the hierarchical CNN layer plus this report (W8).

14 Limitations and Next Steps

- **Leave-one-capture-out evaluation.** The CNN fine-layer (previous section) is built and works end-to-end (91.8 %), but its fine-split scores benefit from capture leakage; a leave-one-capture-out protocol is the deployment-honest way to report generalisation.
- **Flow-level features** are the principled lever for finer attack-family resolution — and would likely help the residual benign↔scan boundary too.
- **Broader evaluation**, per the proposal, is still outstanding: CICIoT2023 [10] (primary), TON_IoT cross-dataset transfer, ET-BERT / XGBoost / CNN-BiLSTM baselines, FGSM adversarial robustness [23], and the int8 Raspberry-Pi edge-latency benchmark (current 5 ms is on an M3 Pro CPU, not edge hardware).
- **Longer-horizon directions.** Per-alert explainability for zero-day alerts [22] and federated cross-site pre-training [24, 25] are natural extensions of the design.

15 Conclusion

Starting from a single-capture prototype, `flowmamba` now runs on seven real IoT-23 malware captures as a three-class gateway detector at **98.2 % accuracy**, macro-F1 **0.984**, anomaly ROC-AUC **0.935**, and ∼5 ms/flow — inside the project's 10 ms budget, at 2 MB. More importantly, the journey produced a defensible *finding*: payload-free flow-patterns resolve attacks by structure, not family, so the honest taxonomy is three classes, and finer labels need richer features rather than a bigger model. That boundary is exactly what the next CNN experiment is designed to probe.

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